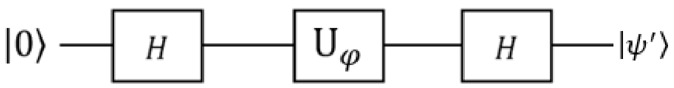
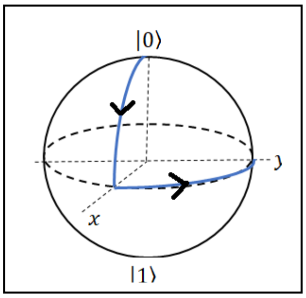
ELK336-Quantum Computation
HOMEWORK #2
Due 09.05.2024

-The students are obligated to do the homework themselves.
-The copied homework will not be considered or graded.
-Homework should be ready before 23.00 on the ninova in electronic form.

1- Given the following quantum circuit ;



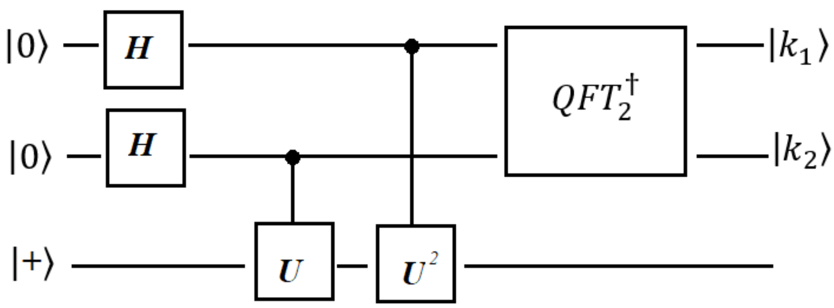
(a) Find the $|\psi'\rangle$ output state. Here $U_\varphi = |0\rangle\langle 0| + e^{i\varphi}|1\rangle\langle 1|$ is a relative phase gate.
(b) Show that the relative phase φ applied by U_φ can be calculated as $\varphi = cos^{-1}[p_0 - p_1]$ where p_0 and p_1 are the probabilities finding the $|\psi'\rangle$ state in $|0\rangle$ and $|1\rangle$ respectively.



2- A quantum state initialized in $|0\rangle$, follows the path in the figure (left panel) and reaches $|0\rangle \rightarrow |\psi'\rangle$.

(a) Which unitary rotations performed for the path? Construct the matrix representations of the respective rotations.
(b) Draw the circuit diagram gives $|\psi'\rangle$ and calculate $|\psi'\rangle$.

3- (a) Given the operator $U = R_x(\pi)$, show that initial state $|+\rangle$ is an eigenstate of U.
Then obtain the phase factor ϕ in the binary form by use of the corresponding eigenvalue $\lambda = e^{2\pi i\phi}$.
Finally show that $|\phi\rangle = |k_1 k_2\rangle = |11\rangle$.
(b) Find the output $|k_1 k_2\rangle$ of the algorithm given below where U is the R_x rotation given above. Draw the whole algorithm including $QFT_2^\dagger$ and obtain $|k_1 k_2\rangle$ step by step (in Dirac notation) starting from $|00+\rangle$.



The questions will be graded as
1-(a) 15pts 1-(b) 15pts, 2-(a)20pts 2-(b)10pts, 3-(a) 20, (b) 20
D.T.